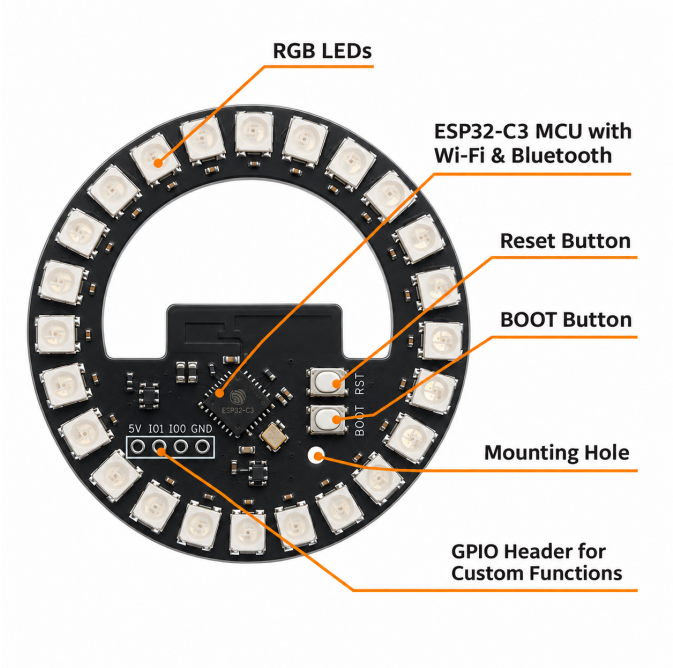
ESP32-C3 RGB LED Ring Development Board

USER MANUAL



Board layout and component identification

1. Product Overview

- Main controller: ESP32-C3 MCU with integrated Wi-Fi and Bluetooth.
- Lighting: 24 onboard WS2812B addressable RGB LEDs arranged in a circular ring.
- Factory demo: Preloaded firmware provides five built-in lighting effects, selectable with the BOOT button.
- Programming: USB Type-C interface supports automatic program downloading; no external programmer is required.
- Expansion: IO0 and IO1 are available for external sensors or communication with another microcontroller.
- Applications: Lighting-effect development, student projects, and ambient lighting.

2. Key Features

Five Built-In Lighting Effects

– Demo firmware is preloaded at the factory.

– Press the BOOT button to switch between five built-in lighting effects.

– The board is ready to use after power-up.

ESP32-C3 Wireless Connectivity

– The onboard ESP32-C3 integrates Wi-Fi and Bluetooth.

– Upload programs through the USB Type-C port using a standard USB data cable.

– No separate programmer is required.

Two GPIO Pins for Expansion

– IO0 and IO1 are available for external connections.

– Connect sensors or communicate with an external microcontroller for further development.

RESET and BOOT Buttons

– RESET restarts the board.

– BOOT switches the demo lighting effects.

– Mounting holes are provided for enclosure installation.

Important: The factory-installed demo firmware does not include app-based remote control. If app connectivity or network-based remote control is required, the app and related firmware functions must be developed by the user.

3. Electrical Specifications

Item	Specification
Main Controller	ESP32-C3
Wireless Connectivity	Wi-Fi + Bluetooth
Power Interface	USB Type-C
Operating Voltage	5 V
Typical Operating Current	200 mA
Exposed GPIO	IO0, IO1 (3.3 V logic)
LEDs	24 × WS2812B addressable RGB LEDs
Board Diameter	6.5 cm
Buttons	RESET button; BOOT button
GPIO Logic Level	3.3 V. Do not connect 5 V signals directly.

4. Documentation & Downloads

Download the files and development tools using the links below.

Resource Archive (7Z):
https://testxinlu.oss-cn-beijing.aliyuncs.com/static/upload/file/warehouse/2026/08/17/1786959176405633.7z

Arduino IDE Download / Resources:
https://www.xinlucity.com/?s=resourcedetail/index/id/117.html

Thonny Download / Resources:
https://www.xinlucity.com/?s=resourcedetail/index/id/118.html

ESP32 Offline Support Package:
https://pan.baidu.com/s/1teLKfDVP9NXphiU6RN3Ecw?pwd=1234

Baidu Netdisk extraction code: 1234

5. Development Software

- Arduino IDE: Use the Arduino resource link above to access the software download or related setup materials.
- Thonny: Use the Thonny resource link above to access the software download or related setup materials.
- ESP32 offline support package: Download the package from the Baidu Netdisk link above and enter the extraction code when prompted.

6. GPIO Voltage Warning

- The exposed GPIO pins use 3.3 V logic. Do not apply 5 V signals directly to IO0, IO1, or other GPIO connections.